Detailed PDC Target-Momentum Reconstruction and Uncertainty Analysis

Methods, Error Models, and Confidence/Credible Intervals

SMSimulator5.5 Analysis

April 20, 2026

Coordinate frames: lab vs. target (beam-as-Z)

Two right-handed frames:

- **Lab frame** (Geant4 world): $+z$ along the nominal SAMURAI beamline. PDC hits, field map, RK integration, and the internal fit live here.
- **Target frame** ("beam-as-Z"): $+z$ along the incident beam; rotated about y by $-\alpha_{\mathrm{tgt}} = -3^\circ$ relative to lab. `truth_proton_p4` stays in this frame (see `DeutPrimaryGeneratorAction.cc`: the forward $R_y(-\alpha)$ is applied to a local copy only; `fMomentum` is never rewritten).

All truth vs. reco plots and all interval coverage numbers in this report are in the target frame. Reco tools apply $R_y(+\alpha_{\mathrm{tgt}})$ at the write stage; the 3×3 momentum covariance is rotated as $\Sigma' = R\Sigma R^{\mathrm{T}}$; $px/py/pz$ Fisher/Laplace intervals are Gaussian-rebuilt from the rotated diagonals; $|\vec{p}|$ is rotation-invariant. MCMC/profile asymmetric quantile information degrades to Gaussian fallback. Prior to 2026-04-19 this rotation was missing; the affected $\Delta p_x \approx -p_z \sin\alpha \approx -33$ MeV/c bias contaminated all pre-fix plots ($p_z \approx 627$ MeV/c; ≈ -52 MeV/c for $p_z \approx 1$ GeV/c). Fisher 68% p_x coverage recovered from ≈ 0.05 to ≈ 0.80 after the fix. Shared helper: `libs/analysis_pdc_reco/include/PDCFrameRotation.hh`.

Study Scope

Physics question

- Reconstruct proton target momentum from a two-point PDC track in the SAMURAI field.
- Quantify not only the central estimate, but also the uncertainty attached to that estimate.

This report focuses on

- Runtime reconstruction backends now exposed by `reconstruct_target_momentum`
- RK-based uncertainty analysis implemented in `analysis_pdc_reco`
- Diagnostic comparison between RK, NN, and multidimensional regression branches
- Confidence and credible intervals for p_x, p_y, p_z, and $|\vec{p}|$

Top-Level Reconstruction Methods

Method	Input / model	Current strength	Current limitation
RK fit	Field-map back-tracing from target to PDC constraints	Physics-based, interpretable, supports uncertainty propagation	Slowest branch; quality depends on initialization and fit mode
NN regressor	Direct map from PDC coordinates to (p_x, p_y, p_z)	Best point-estimate accuracy on current truth grid	No interval output in runtime yet
MultiDim surrogate	Polynomial surrogate from offline regression	Fast when calibrated	Current truth-grid sample has no matched events; no interval support

The current uncertainty study is therefore centered on the RK branch, because it is the only branch that now exposes Fisher, profile-likelihood, Laplace, and MCMC interval estimates.

RK Reconstruction Modes

The RK backend currently supports three fit modes

1. **two-point-backprop**: infer momentum by back-propagating the PDC line to the target in the field map.
2. **fixed-target-pdc-only**: keep the target anchor fixed and fit the momentum state against the two PDC points.
3. **three-point-free**: fit both target point and momentum state as free parameters under target and PDC constraints.

Why this matters for uncertainty

- The parameterization determines the Hessian conditioning and therefore the Fisher covariance.
- The fit mode also changes the profile-likelihood geometry and the posterior shape.
- In the present implementation, the enlarged legacy-run analysis uses `fixed-target-pdc-only`, because it is the most stable mode for real-style reconstructed input.

What Is Written by the RK Error Analysis

Runtime ROOT branches now available

- Central fit diagnostics: status, method, iterations, raw and reduced χ^2, N_{dof}
- Fisher widths: σ_{p_x}, σ_{p_y}, σ_{p_z}, $\sigma_{|\vec{p}|}$
- Optional Laplace intervals: 68% and 95% lower/upper bounds for p_x, p_y, p_z, and $|\vec{p}|$

Offline diagnostic tool

- `build/bin/analyze_pdc_rk_error`
- Outputs: `track_summary.csv`, `profile_samples.csv`, `bayesian_samples.csv`, `summary.csv`, `validation_summary.csv`, and summary plots
- This tool re-runs the RK fit on reconstructed PDC tracks and computes frequentist and Bayesian intervals on sampled candidates

Interval Definitions

Estimate	Construction	Statistical meaning	Current use
Fisher	Inverse local Hessian around the RK optimum	Local Gaussian approximation to the likelihood	Fast width estimate for all successful tracks
Profile likelihood	Scan one fit parameter, minimize others, use $\Delta\chi^2 = 1, 4$ crossings	Frequentist confidence interval	Asymmetry and nonlinearity diagnosis on sampled tracks
Laplace posterior	Gaussian posterior near the mode using posterior covariance	Bayesian credible interval	Fast Bayesian interval for all tracks if posterior is valid
MCMC posterior	Random-walk samples from the posterior	Bayesian credible interval with shape diagnostics	Validation on sampled tracks, not on every event

Important distinction: Fisher and Laplace are local approximations; profile likelihood and MCMC are more robust to non-Gaussian structure but much slower.

Synthetic Truth-Grid Method Comparison

Diagnostic dataset

- Existing five-point truth grid around $p_z = 627$ MeV/c
- One matched event per truth point in the current regression fixture
- Use this only as a backend sanity check, not as a final performance benchmark

Backend	Matched events	Mean $\lvert\mathrm{bias}_{p_x}\rvert$	Mean $\lvert\mathrm{bias}_{p_y}\rvert$	Mean $\lvert\mathrm{bias}_{p_z}\rvert$
NN (contaminated model)	5	21.04	15.56	3.10
RK fixed-target	5	178.35	11.46	234.39
RK three-point	5	651.35	15.77	190.31
RK two-point	5	67.99	1.40	604.92
MultiDim	0	n/a	n/a	n/a

On this fixture, the NN branch is currently the strongest point estimator, while the RK modes still need calibration or configuration tuning for this truth-grid operating point. Audit note 2026-04-20: the NN row was produced by the deleted polluted model; the clean retrain is expected to keep (and likely extend) its point-estimate lead, especially on p_x, p_y. Also note RK numbers here predate the 2026-04-16 RK bugfix — see `rk_reconstruction_status_20260416` for current RK

How to Read the Truth-Grid Result

Interpretation

- The truth-grid test is very small, so the table is dominated by pointwise behaviour rather than stable statistics.
- Even so, the pattern is useful: each RK mode fails differently.
- `two-point-backprop` keeps transverse components relatively close, but collapses p_z.
- `fixed-target-pdc-only` is more balanced and therefore chosen as the first uncertainty-analysis mode for legacy reconstructed samples.
- `three-point-free` is the most flexible and also the least stable on the current small truth-grid fixture.

Practical conclusion

- Use the NN branch when the immediate goal is best point prediction.
- Use the RK branch when the goal is uncertainty-aware reconstruction and physically interpretable interval estimates.

Expanded Legacy Reconstruction Sample

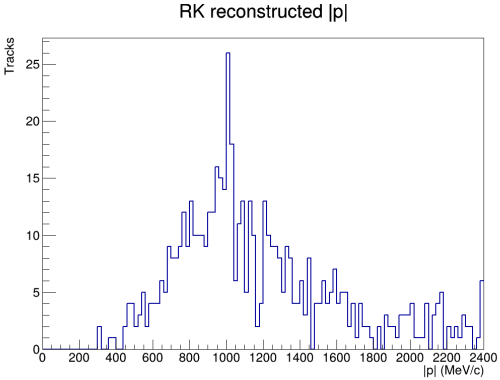
Input class

- Legacy reconstructed `after_pdcana` ROOT files
- Geometry macro: `geometry_B115T_pdcOptimized_20260227.mac`
- Magnetic field map: `180703-1,15T-3000.table`, dipole rotation 30°
- RK mode: `fixed-target-pdc-only`

Expanded run configuration

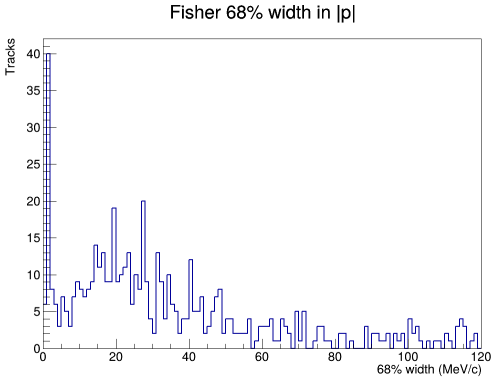
- Up to 500 events per file
- Profile scan: 11 points per parameter, 4 sampled tracks per momentum quartile
- Bayesian validation: 2 sampled tracks per quartile, 120 kept MCMC samples after burn-in/thinning
- Current completed output: 815 successful RK tracks, 32 profile samples, 16 Bayesian samples

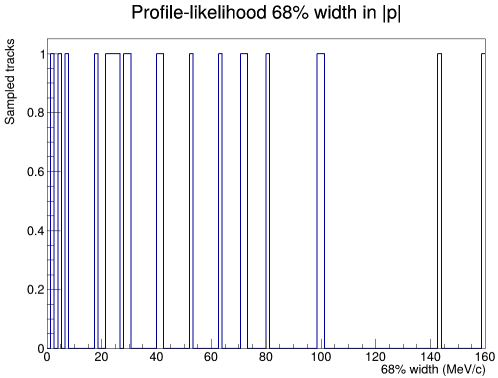
Reconstructed Momentum Spectrum



The enlarged legacy sample covers a broad reconstructed proton momentum range. The median $|\vec{p}|$ is 1370.4 MeV/c with interquartile range 970.8 to 2920.8 MeV/c.

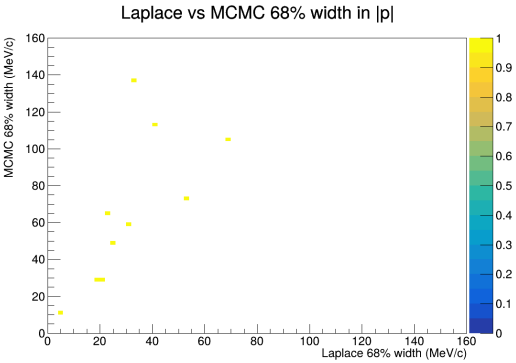
Fisher and Profile-Likelihood Widths





In the enlarged run, the median 68% width in $|\vec{p}|$ is 48.20 MeV/c from the Fisher estimate and 76.91 MeV/c from profile likelihood. The profile interval remains substantially broader, so the likelihood is still not well described by a purely local Gaussian approximation.

Laplace Versus MCMC



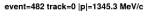
Enlarged run

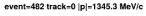
- Median Laplace 68% width in $|\vec{p}|$: 48.14 MeV/c
- Median MCMC 68% width in $|\vec{p}|$: 109.32 MeV/c
- Median MCMC acceptance rate: 0.593
- Median effective sample size: 9.23

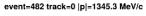
Reading

- Laplace tracks Fisher closely, as expected for two local Gaussian approximations.
- MCMC is much broader on sampled tracks, which again suggests non-Gaussian posterior structure.

Representative Profile Curves







This is the clearest frequentist diagnostic. Symmetric parabolic curves would support Fisher-like treatment. Asymmetry, flat directions, or clipped crossings explain why profile-likelihood intervals can be wider than Hessian-based widths.

Confidence and Credible Interval Summary

Method	Tracks covered now	68% interval type	Main message
Fisher	All successful RK tracks	Local confidence width	Good for speed, optimistic when the likelihood is curved or asymmetric
Laplace	All successful RK tracks with valid posterior covariance	Local Bayesian credible interval	Similar to Fisher in the current smoke study
Profile likelihood	Sampled RK tracks	Frequentist confidence interval	Captures asymmetry and broader nonlocal structure
MCMC	Sampled RK tracks	Bayesian credible interval	Best shape information, but highest runtime cost

For current PDC legacy reco data, the safest interpretation is: Fisher/Laplace for fast per-track monitoring, profile/MCMC for method validation and interval realism checks.

Current Limitation on Coverage Validation

What is missing

- The legacy `after_pdcana` files contain an `OriginalParticleData` branch without a safe dictionary in this workflow.
- Direct truth binding from those files caused the earlier memory-corruption crash and has been disabled in the analysis tool.

Consequence

- We can compute interval sizes and compare methods on reconstructed tracks.
- We cannot yet claim empirical 68% or 95% coverage on the legacy sample.
- Coverage validation must be done on simulation files with a safe truth interface or by adding a reliable dictionary-backed legacy reader.

Implementation Status

Completed

- RK runtime branches for central fit diagnostics and interval outputs
- Offline tool for Fisher, profile-likelihood, Laplace, and MCMC comparison
- Synthetic backend comparison CSV summaries
- Stability fix for the legacy-file crash path caused by unsafe truth-branch binding

Still needed

- Calibrate RK modes against larger truth-labeled samples
- Add interval support for the NN branch, for example heteroscedastic regression or quantile targets
- Restore truth-based coverage validation for legacy-style files through a safe reader path

Takeaway

- The PDC reconstruction framework now supports several central-estimate methods, but only the RK branch currently supports rigorous uncertainty analysis.
- On the available legacy reconstructed sample, local Gaussian intervals are much narrower than profile-likelihood and MCMC intervals, so Gaussian-only error bars are not sufficient.
- The NN branch remains the strongest point predictor on the current truth-grid diagnostic, but it is not yet uncertainty-aware.
- The next technically important step is not another point-estimate tweak; it is truth-backed interval validation on a larger, stable dataset.